Mediconnect AI:A RAG based medical chatbot

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***Abstract*—In this study, we demonstrate an innovative use of the Retrieval-Augmented Generation (RAG) architecture to improve patient care and simplify medical consultations. We successfully converted audio conversations into text by using OpenAI's Whisper model. The GPT-4.0 model was then used to extract the information from these transcripts into insights that could be put to use. Vector embeddings of the condensed text were created and saved in the Pinecone vector database to enable effective data retrieval. Cosine similarity metrics-based retrieval chains were used to dynamically retrieve relevant information.We also incorporated this pipeline into an easy-to-use chatbot application that was constructed using Streamlit. The chatbot's two main features—a Symptom Checker for first diagnosis and a Patient Insights tool for condensing important health data—significantly lessen doctors' administrative burdens while providing patients with early diagnostic assistance. This holistic approach demonstrates the potential of RAG architectures in transforming healthcare data workflows and enhancing decision-making processes in clinical settings.**

***Index Terms*—Embeddings, Tokenization, RAG, Vector stores, Cosine similarity, Open AI, ChatGPT.**

I. Introduction

Growing demand and wider profit margins are driving a significant transition in the healthcare sector. This industry, which is expected to increase from $583billionin2022to819$ billion by 2027, has significant obstacles that affect provider efficiency and patient care. The excessive administrative burden that takes up healthcare professionals' valuable time is a major problem.

A. Extensive time on patient histories:

Providers have limited time for direct patient treatment because administrative duties like reviewing and keeping patient records take a lot of time and sometimes include complex documentation systems and legal requirements. In addition to having an impact on treatment quality, this also makes healthcare professionals more likely to experience burnout and discontent with their jobs.

B. Long waiting periods

In the healthcare industry, lengthy wait times for appointments have become a major problem. Patients sometimes must wait weeks or even months to see a medical practitioner. serious consequences may result from these delays, especially for those who require prompt medical assistance for acute or chronic diseases.

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C. Solution

In order to help healthcare practitioners make well-informed decisions, an AI-driven technology gives simplified data and immediate preliminary insights into patient symptoms, and their previous conversation summary. The chatbot from Mediconnect AI also help with the problem of long waiting period by offering immediate first evaluations. Medi connect AI ensures that urgent health concerns are promptly handled by incorporating AI into the first patient engagement. This helps with triaging cases and guiding patients through their next steps. This lessens patient worry and avoids the possible worsening of medical issues that might happen if prompt attention is not provided.

So, at the end of the paper, we will explain how we built a fully functional medical chatbot using audio conversations between doctors and patients and implementation of RAG architecture.

II. Review of Literature

A. The need for Audio conversion to text:

The widespread adoption of electronic health record (EHR) systems has significantly increased the administrative workload for healthcare professionals. Doctors now spend approximately six hours of their 11-hour workdays using EHR systems, with 1.5 hours dedicated solely to documentation tasks (Shanafelt et al., 2015). This extensive administrative burden contributes to burnout among physicians, further exacerbated by a growing shortage of primary care providers. Automatic Speech Recognition (ASR) technology offers a promising solution by automating the transcription of clinical visits, allowing doctors to focus more on patient care. Advanced models like Connectionist Temporal Classification (CTC) and Listen, Attend, and Spell (LAS) have demonstrated efficacy in speech recognition tasks, achieving word error rates (WER) of 20.1% and 18.3%, respectively (Chan et al., 2016). However, these models have limitations in handling real-world noise and diverse speech patterns, creating a need for more robust solutions like Whisper AI, which is better suited for clinical environments due to its scalability and efficiency.

B. Importance of Text Summaries:

With the rise of patient-centered healthcare models and the expansion of remote consultation platforms, summarized text data has become vital. Platforms like "Internet Plus

Healthcare" allow doctors to provide remote consultations, enhancing efficiency and reducing time spent on repetitive tasks while maintaining personalized care (Zhou et al., 2024). Summarizing audio conversations into concise text benefits both physicians and patients by offering rapid access to key information, improving care quality, and fostering knowledge exchange. Extractive and abstractive summarization algorithms play a crucial role by condensing complex medical dialogues and EHR data into easily digestible formats(Chen 'I& Lee,2023). These summaries enable healthcare providers to make timely, informed decisions and ensure patients receive comprehensive yet accessible insights into their care.

C. Why use AI for Text Summaries:

From its usage in brain biopsy trajectory mapping to its current applications in hospital logistics and medicines, artificial intelligence (AI) has revolutionized the healthcare industry since the 1980s (Khan 'I& Anwar, 2019). The drawbacks of previous ASR systems are addressed by AI-driven speech-to-text models such as Whisper AI, which provide excellent transcription accuracy and efficiency even in noisy, real-world settings. Whisper AI is a great option for transcribing doctor-patient talks because of its end-to-end architecture and ability to interpret many languages (Radford et al., n.d.). In contrast to conventional techniques, AI models seek to improve transcribing efficiency and quality rather than completely replace human intervention (Bokhove 'I& Downey, 2018).

D. Reasons RAG Models Are Important:

Retrieval-Augmented production (RAG) models, which incorporate external input into real-time content production, are a breakthrough in artificial intelligence. For summarizing dynamic and changing medical discussions, RAG models are especially useful since they obtain important external information to supplement text, unlike typical large language models (LLMs) that only use pre-trained data (Toukmaji 'I& Tee, 2024). RAG models improve medical summaries' relevance and accuracy by fusing the advantages of generation and retrieval, guaranteeing that healthcare professionals have access to current, useful information. RAG models are positioned as essential instruments for enhancing healthcare decision-making and operational effectiveness because to their features.

III. Objectives of Study:

- Reduce Administrative Burden: Create AI-powered tools to automate the transcription and summarizing of doctor-patient talks, reducing the need for manual documentation.
- Improve Care Quality: By cutting down on time spent on administrative duties, healthcare personnel can focus more of their attention to patient care.
- Enhance Decision-Making: Make use of AI models such as Whisper and RAG to deliver precise and up-to-date information for prompt and well-informed decisions in medicine.

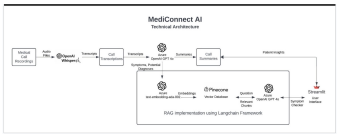

Fig. 1. Mediconnect AI Architecture

- Encourage Efficiency: To solve issues like lengthy wait times and delayed diagnosis, streamline healthcare workflows with cutting-edge AI technology.

IV. Data Collection:

Medical interviews taken during Objective Structured Clinical Examinations (OSCE) make up the dataset. With an emphasis on diagnostic and evaluation procedures, these recordings replicate actual interactions between doctors and patients. The collection is intended to aid in the creation of AI-powered tools for medical conversation analysis, summarization, and transcription.

Contains 272 simulated cases of medical interactions between physicians and patients. Each case is classified into one of the following five medical categories: Respiratory Diseases (majority of cases). Musculoskeletal Diseases. Cardiac Diseases. Dermatological Diseases. Gastrointestinal Diseases. Data Contents:

Each simulation captures detailed physician-patient interactions and cleaned transcripts for reference. Focuses on various medical conditions and their diagnostic assessments.

Click here for Dataset

V. Methodology and Implementation:

We have built the architecture and implemented our workflow in stages.

*A. **Stage 1:***

In this stage, OpenAI Whisper Base Model was used to transcribe audio interactions between patients and doctors, providing a reliable automatic transcription solution. The "text-embedding-ada-002" model was used to create embeddings of the cleaned and transcribed script data to validate these transcriptions. Cosine similarity, which calculates the cosine of the angle between two non-zero vectors, was used to evaluate the correctness of these embeddings. According to Han et al. (2011), these metric measures how similar two text representations are to one another; a cosine similarity score nearer 1 denotes more alignment.

A one-sample t-test was used to statistically assess the transcribing accuracy. The test established a significance level (α) of 0.05 and compared the transcriptions' cosine similarity scores to a 95% population mean. The null hypothesis was rejected because of the computed t-statistic and one-tailed p-value, indicating that Whisper's transcribing accuracy is higher than 95%.

- **Lang Chain Framework Integration**:

The LangChain framework was used to improve communication with OpenAI's language models. To build up the

setup, libraries like langchain_openai, langchain_core, and langchain_community were installed. To facilitate querying and evaluating transcription results, embeddings were developed using the text-embedding-ada-002 approach.

"From langchain.embeddings.openai import OpenAIEmbeddings

embed = OpenAIEmbeddings(model='text-embedding-ada-002', openai_api_key='sk-proj') "

By combining OpenAI's LLM with transcription validation, embedding analysis, and interactive features, this method streamlines the process for audio-to-text conversion and assessment. **NOTE**: We have used cloud based OpenAI LLM and hence we are not revealing our API key here after because of computational cost and security reasons.

B. Stage 2:

In order to extract important information from cleaned transcripts of patient-provider interactions, the second phase concentrated on text preparation and summarization. The transcripts were processed using GPT-40-mini to provide structured summaries, which allowed for effective analysis. In particular, the model was instructed to separate the discussion into four distinct parts: Key patient information: that was mentioned during the session was extracted. Symptoms Discussed: The patient's claimed symptoms were identified. Diagnosis and Assessment: Provided an overview of the condition's diagnosis and evaluation by the medical professional. Treatment Plan: Describes the drug or treatment plan that has been prescribed. The CSV file containing Audio file ID and the summarized data produced by this method is then utilized by our chatbot to build patient insights features.

- **Implementation**: "def summarize_transcipt(transcript: str) str: chat_completion = client.chat.completions.create(messages=["role": "system", "content": "'Read the following call transcript between a provider and a patient, generate a detailed summary in 4 following sections: Patient Information: Key details about the patient discussed in the call. Symptoms Discussed: A list of symptoms the patient reported. Diagnosis and Assessment: The provider's assessment and diagnosis of the patient's condition. Treatment Plan: Details of the treatment plan or medications prescribed. "', "role": "user", "content": transcript], model="gpt-4o-mini", max_tokens=1000) return chat_completion.choices[0].message.content"

This method returns a structured summary after receiving a transcript as input. Because of its sophisticated ability to handle complex medical talks, the GPT-40-mini model was selected. The model's output was stored in a CSV file, guaranteeing a structured format for incorporating patient insights feature in the chat bot development.

C. Stage 3:

Part-1: To ensure accuracy and consistency, the data was preprocessed to identify the key points prior to vector storage and Retrieval-Augmented Generation (RAG) implementation. A structured DataFrame containing the retrieved data was created and used as the RAG architecture's reference data. Python workflow: To process each transcript using the intended prompt and extract the required data, a Python program was put into place. For simple connection with the OpenAI API, the function makes use of the LangChain framework.

- **Prompt Template**: "prompt_template = PromptTemplate(input_variables=["data"], template="'Assistant, I need you to analyze the transcript between a doctor and a patient and extract specific information. Your task is to give the Reason for Call::...¡/c¿, Age:...¡/a¿, Gender:...¡/g¿, Symptoms:...¡/s¿, Disease:...¡/d¿, Diagnosis:...¡/di¿, and Prescription:...¡/p¿ based on the conversation.
  Here is the transcript: data Please extract the necessary information and including all the required details. Remember to accurately identify and always in this format Reason for Call:..¡/c¿, Age:...¡/a¿, Gender:...¡/g¿, Symptoms:...¡/s¿, Disease:...¡/d¿, Diagnosis:..¡/di¿, and Prescription:..¡/p¿. If not interpretable, not specified, not mentioned for example in age, Age:NA¡/a¿ so on. Text: "') chat_model = ChatOpenAI(api_key=openai.api_key,model="gpt-4o",temperature=0.0)
  chain = LLMChain(llm=chat_model, prompt=prompt_template) "

Part-2:

This section describes the entire process of putting into practice a Retrieval-Augmented Generation (RAG) system that uses Pinecone, LangChain, and vector databases to create a reliable diagnosis assistance. Using a pre-processed dataset, the system analyzes user input (symptoms) to provide pertinent diagnoses, likely illnesses, and medicines.

- 1.**Creation of Embedding**: To store and retrieve textual data in a vectorized format: Embedding Model: To ensure semantic comprehension of the input, high-dimensional vector representations of textual data were created using OpenAI's text-embedding-ada-002 model.
- 2.**Configuring a Vector Database**: Pinecone was chosen as the database because of its speed and scalability in handling vectorized data. Making an Index: With the following requirements, a new index called disease-symptoms-gpt-4 was developed. *Implementation*:("pc.create_index(name="disease-symptoms-gpt-4", dimension=len(docs['VectorSummary'][0]), metric="cosine", spec=ServerlessSpec(cloud="aws", region="us-east-1")
- 3.**Ingestion of Data** The Pinecone vector database was fed the pre-processed dataset, which included diagnoses, illnesses, medications, and summaries: A unique identity, the vector representation, and metadata (such as Summary, Disease, Diagnosis, and Prescription) were added to each entry before it was upserted into the database.
  Contextual retrieval during query processing was guaranteed via metadata.
  Implementation: ("for i, j in docs.iterrows():

index.upsert(vectors=[(j['Unnamed: 0'], j['VectorSummary'], 'text': j['Summary'], 'Disease': j['Disease'], 'Diagnosis': j['Diagnosis'], 'Prescription': j['Prescription'])])

- 4.**Integration of Vector Store with LangChain**: LangChain was updated to use the Pinecone vector storage for retrieval purposes:
  The Pinecone module of LangChain was used to cover the vector storage for query interaction.
  Embedding Query: To convert queries into vectors for comparison with the stored vectors, the same embedding model was used.
  Implementation: from langchain.vectorstores import Pinecone vectorstore = Pinecone(index, embed.embed_query, "text")
- 5.**Prompt Template:** For the retrieval-augmented diagnostic system, a specially created prompt template was created. It provided the background knowledge and directed the model to provide precise findings using the data that was collected.
  Prompt Template: When symptoms are entered into the system, related illnesses, diagnoses, and medicines are retrieved. The system responds politely in the event that data is unavailable. *Implementation:* prompt_template = "' Accept the user's symptoms as input and provide probable diseases, diagnoses and prescription using only the information stored in the vector database. If the information is not available in the database, politely inform the user that the data is insufficient to provide a diagnosis. Text: context "' PROMPT = PromptTemplate(template=prompt_template, input_variables=["context"]) ")
- 6.**QA chain for Retrieval:** The VectorStoreRetriever in LangChain was used to develop a retriever. With the retriever and ChatGPT-4o as the LLM for producing the answer, a RetrievalQA chain was started. *Implementation:* retriever = VectorStoreRetriever(vectorstore=vectorstore) qa_chain = RetrievalQA.from_chain_type(llm=ChatOpenAI(api_key="sk-proj-...", model_name='gpt-4o', temperature=0.0), chain_type="stuff", retriever=retriever, chain_type_kwargs="prompt": PROMPT, ")

End-to-End workflow: Input Processing: The system receives user symptoms. Vectorization: To compare semantic similarity, the input is embedded using the same text-embedding-ada-002 model. Retrieval: Using cosine similarity, the vectorized query extracts the most pertinent entries from the Pinecone vector database. Response Generation: The LLM receives the created prompt and the context that was obtained. If there is not enough information, the model either produces a diagnosis and prescription or offers an alternate response.

D. *Stage 4:*

The creation of the "MediConnect AI" interactive chatbot interface is the main goal of this phase. We have used stream lit interface to build the chatbot and the main objective is to give consumers an easy-to-use platform for assessing symptoms and retrieving patient insights from earlier discussions.

VI. Results

Our results and core work are simply defined by our chatbot. Please feel free to use and explore it.

Link :Mediconnect AI

VII. Conclusion

A. *Custom LLM Development:*

There is a great deal of promise to improve accuracy and stability by creating a customized large language model (LLM) for use cases in healthcare. Better management of complex medical language, complicated settings, and particular difficulties unexpected to the healthcare industry is made feasible by refining the LLM using bigger medical datasets. The model is especially well-suited to situations where medical language and context are crucial because of this modification, which guarantees that it can read domain-specific data.

Such a model's seamless data sharing with Electronic Health Records (EHR) and other healthcare IT systems allows for real-time insights and supports healthcare professionals' decision-making. In the end, this strategy seeks to empower doctors by giving them AI-driven insights that increase their capacity to provide individualized care and boost patient outcomes.

B. *Healthcare Empowerment and Redefining Patient Care:*

The LLM has the potential to revolutionize patient care by utilizing AI-driven insights to change the way medical practitioners engage with data and make clinical judgments. By providing individualized therapy alternatives based on each patient's needs, these models can close the gap between raw healthcare data and workable treatments. The objective is to expedite the use of AI in clinical settings by equipping healthcare professionals with the tools they need to make well-informed choices fast. In addition to increasing the effectiveness of healthcare delivery, this change has the potential to completely redefine patient care, leading to better results and increased satisfaction.

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